

CARTAN MOVING FRAMES AND DISCRETE EXTERIOR CALCULUS FOR SELF-SUPERVISED PROTEIN STRUCTURE REPRESENTATION

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ABSTRACT

Masked autoencoding over $SE(3)$ -invariant residue frames is an effective recipe for self-supervised protein structure representation, yet it treats a protein backbone as a bag of local geometric features and leaves the *intrinsic* differential geometry of the structure — how local frames transport and twist relative to one another — largely implicit. We introduce **Cartan-DEC-MiAE**, which recasts structural masked autoencoding in the language of *Cartan moving frames* and *discrete exterior calculus* (DEC). Each residue is a moving frame; edges carry discrete connection 1-forms, triangles carry curvature 2-forms, and closed loops carry a holonomy defect. A lightweight *Cartan–Darboux* decoder predicts these gauge-invariant forms for masked residues and analytically integrates them back to 3D geometry, so the encoder is forced to internalise transport-aware structure. We further inject curvature and connection-heat-transport biases into the sparse encoder attention. Under a strictly matched from-scratch protocol on the full 369,740-structure TED-Bench corpus, the Cartan-DEC objective improves top-1 fold-classification accuracy by +2.4 (in-distribution) and +10.7 (external CATH 4.4) and macro F1 by +5.5/+11.0 over an identically trained vanilla MAE; curvature-biased attention adds a further +3.3/+13.0 top-1. Extensive controlled ablations dissect which geometric invariants matter: curvature 2-form supervision is the single most valuable loss term, combining curvature with connection-heat attention prevents attention collapse on sparse inputs, and — as a cautionary result we report faithfully — holonomy-targeted masking does not beat simple contiguous masking, and higher reconstruction fidelity *anti-correlates* with representation transfer. Our results position discrete differential geometry as a principled and practically useful inductive bias for structural self-supervision.

1 INTRODUCTION

Protein function is dictated by three-dimensional structure, and the explosive growth of predicted structures — hundreds of millions of models in the AlphaFold Database (Jumper et al., 2021; Varadi et al., 2022) — has turned structure-based representation learning into a central problem in computational biology. A canonical probe of whether a model captures global structural organisation is *fold classification*: assigning a domain to one of the hundreds of CATH topologies (Sillitoe et al., 2021). The recent TEDBench benchmark (Lau et al., 2024), with its long-tailed distribution over 965 topology classes, makes this an unusually stringent test of structural understanding.

A strong and scalable recipe for this setting is the *masked invariant autoencoder* (MiAE), which ports the masked autoencoding paradigm (He et al., 2022) to protein backbones: each residue is represented by a local $SE(3)$ frame (Ingraham et al., 2019; Hayes et al., 2025), a large fraction of frames is masked, a heavy encoder processes only the visible residues, and a light decoder reconstructs backbone geometry. This is efficient and effective, but the learning signal is fundamentally *local*: the model reconstructs per-residue positions and directions, treating the backbone as a collection of rigid pieces. The way those pieces *fit together* — how one frame is transported into the next, how curvature accumulates around a β -barrel, how a domain hinge introduces a topological “twist” that no local descriptor can express — is exactly the information a fold classifier needs, and it is only implicitly available in a coordinate-reconstruction objective.

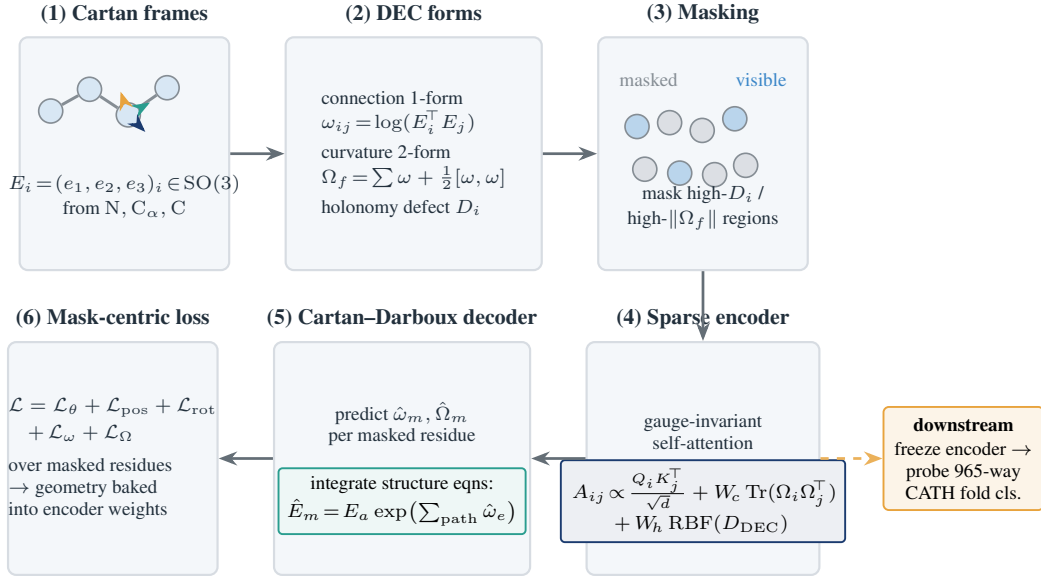


Figure 1: Overview of Cartan-DEC-MiAE. (1) Each residue is a Cartan moving frame $E_i \in \text{SO}(3)$ built from backbone N, C_α , C atoms. (2) Discrete exterior calculus turns the frame field into gauge-invariant forms: connection 1-forms $\omega_{ij} = \log(E_i^\top E_j)$ on contact edges, curvature 2-forms Ω_f on contact triangles, and a per-residue holonomy defect D_i . (3) A large fraction of residues is masked. (4) A sparse, gauge-invariant encoder processes only the visible residues, with attention biased by curvature inner products and connection heat-transport. (5) A Cartan–Darboux decoder predicts the forms for masked residues and integrates the discrete structure equations back to 3D geometry. (6) A mask-centric loss over all differential forms bakes transport-aware geometry into the encoder weights, which are then frozen and probed for 965-way fold classification.

We argue that the right language for this missing structure is classical differential geometry in its *discrete* form. Cartan (1983)’s method of moving frames describes a manifold not by coordinates but by how an orthonormal frame moves along it, encoded in connection and curvature forms; discrete exterior calculus (DEC) (Desbrun et al., 2005; Crane et al., 2013) provides a faithful, computable discretisation of these objects on meshes and graphs. We bring this machinery to protein self-supervision. Our method, **Cartan-DEC-MiAE** (Figure 1), models a backbone as a discrete differential manifold: residue frames $E_i \in \text{SO}(3)$, discrete connection 1-forms $\omega_{ij} = \log(E_i^\top E_j)$ on the contact graph, curvature 2-forms Ω_f on contact triangles, and a per-residue *holonomy defect* D_i measuring frame-transport frustration around closed loops. A *Cartan–Darboux* decoder is trained to predict these forms for masked residues and to integrate the discrete Cartan structure equations back into frames and coordinates; because the targets are gauge-invariant and reconstruction requires long-range composition, the encoder is pushed to learn transport-aware structure rather than to interpolate neighbours.

We make the strongest possible empirical case for this objective by holding *everything else fixed*. On the full 369,740-structure TEDBench-train corpus, we pretrain an identical encoder from scratch under a vanilla MAE objective and under the Cartan-DEC objective, and probe both with an identical frozen-encoder pipeline. The DEC objective improves top-1 accuracy by +2.4 on the in-distribution test set and +10.7 on the external CATH 4.4 set, and macro F1 (over all 965 classes) by +5.5 and +11.0; adding a curvature-biased attention prior widens the external gap to +13.0 top-1. A calibration row using the public MiAE-B encoder under our exact probe reproduces its published numbers, confirming the pipeline is faithful.

Because a method built on this much geometric machinery invites the question “*which parts actually matter?*”, we run five controlled ablations that vary one factor at a time. They yield both positive design rules and honest cautionary results. **Positive:** curvature 2-form supervision is the most valuable single loss term (removing it hurts most), and jointly biasing attention with curvature inner products and connection heat-transport is the strongest attention variant and best resists attention collapse

under aggressive masking. **Cautionary:** holonomy-defect-targeted masking — a centerpiece of the original idea — does *not* beat plain contiguous masking at this scale, and, strikingly, the decoder configuration with the *lowest* reconstruction error produces the *worst* downstream encoder. Reconstruction fidelity and representation quality pull in opposite directions, and harder reconstruction targets make better encoders.

Contributions. (i) We formulate protein structural self-supervision in the language of Cartan moving frames and discrete exterior calculus, with a Cartan–Darboux decoder that predicts and integrates gauge-invariant differential forms (§3). (ii) Under a strictly matched full-corpus protocol, we show the Cartan-DEC objective and its curvature-biased attention substantially outperform an identically trained vanilla MAE on 965-way fold classification, in- and out-of-distribution (§4.2). (iii) Through five controlled ablations we identify which geometric invariants drive the gains, and we report — rather than hide — two counter-intuitive negative results about holonomy masking and the reconstruction–transfer trade-off (§4.3).

2 RELATED WORK

Masked autoencoding. Masking-based self-supervision is dominant across modalities: BERT for text (Devlin et al., 2019), and masked autoencoders (MAE) for vision (He et al., 2022), which showed that a heavy encoder on a small visible subset with a light decoder and very high masking ratios learns strong representations. MiAE ports this asymmetric design to protein backbones over SE(3)-invariant frames. Our work keeps the asymmetric MAE skeleton but replaces its coordinate-centric reconstruction target with a differential-form target, and its uniform masking and plain attention with geometry-aware variants.

Protein representation learning. Sequence language models such as ESM (Rives et al., 2021; Lin et al., 2023) learn from amino-acid sequences alone; structure-aware models incorporate 3D geometry through inverse folding (Dauparas et al., 2022; Yang et al., 2023), geometric vector perceptrons (Jing et al., 2021), structure-token vocabularies (Su et al., 2024), or all-atom generative pretraining (Hayes et al., 2025). General E(3)-equivariant networks (Geiger & Smidt, 2022; Batatia et al., 2022; Aykent & Xia, 2025) provide strong supervised backbones. These methods reconstruct or predict *coordinates, distances, or tokens*; none supervise the intrinsic connection and curvature forms of the structure, which is our central proposal.

Discrete differential geometry. Cartan’s moving frames (Cartan, 1983; do Carmo, 1976) and discrete exterior calculus (Desbrun et al., 2005; Crane et al., 2013) give a principled, computable calculus of connection and curvature on discrete domains, underpinning tools such as the vector heat method (Sharp et al., 2019). Proteins are a natural fit — a backbone is a framed discrete curve embedded in 3D with a rich contact graph — yet, to our knowledge, these objects have not been used as *self-supervision targets* for structural representation learning. Cartan-DEC-MiAE makes the connection 1-forms, curvature 2-forms, and loop holonomy the quantities a network must predict and integrate.

3 METHOD

3.1 PRELIMINARIES: CARTAN MOVING FRAMES

Following standard frame-based protein modelling (Ingraham et al., 2019; Hayes et al., 2025), we attach to residue i a rigid frame $T_i = (E_i, t_i) \in \text{SE}(3)$, where $t_i \in \mathbb{R}^3$ is the C_α position and $E_i = (e_1, e_2, e_3)_i \in \text{SO}(3)$ is an orthonormal frame built from the backbone N, C_α , C atoms. The frame field $\{E_i\}$ is Cartan’s *moving frame*: rather than describe the backbone by absolute coordinates, we describe how the frame changes from residue to residue, a description that is manifestly invariant to global rigid motions.

3.2 DISCRETE EXTERIOR CALCULUS ON THE CONTACT GRAPH

We build a residue contact graph $G = (V, E)$ with edges e_{ij} whenever $\|t_i - t_j\| < r_c$ (default $r_c = 8 \text{ \AA}$), and a 2-skeleton of contact triangles F . On this discrete manifold we define gauge-invariant differential forms.

Connection 1-forms. The relative rotation between adjacent frames is the discrete connection 1-form

$$\omega_{ij} = \log(E_i^\top E_j) \in \mathfrak{so}(3), \quad (1)$$

the $\mathfrak{so}(3)$ -valued “amount of turning” required to transport frame i onto frame j . The local soldering form $\theta_i = E_i^\top(t_j - t_i)$ expresses displacement in the local frame.

Curvature 2-forms and holonomy. For a contact triangle $f = (i, j, k)$ the discrete curvature 2-form is the exterior derivative of the connection,

$$\Omega_f = \omega_{ij} + \omega_{jk} + \omega_{ki} + \frac{1}{2}[\omega_{ij}, \omega_{jk}] \in \mathfrak{so}(3), \quad (2)$$

measuring how much a frame fails to return to itself after transport around f ; it captures local tertiary packing curvature. Globally, for a closed loop γ the path-ordered *holonomy* $H(\gamma) = \prod_{e \in \gamma} \exp(\omega_e) \in \text{SO}(3)$ quantifies transport frustration, and the per-residue *holonomy defect energy* $D_i = \sum_{\gamma \ni i} \|H(\gamma) - I\|_F$ localises hinges, active-site clefts, and loop crossovers — geometry that is invisible to purely local descriptors.

3.3 CARTAN-DEC MASKED AUTOENCODER

Cartan-DEC-MiAE keeps the asymmetric MAE skeleton (heavy encoder on visible residues, light decoder on the full set) but replaces its target, its attention, and its masking with geometry-aware versions.

Gauge-invariant sparse encoder. The encoder processes only visible residues. Beyond standard geometric self-attention, we add two additive logit biases derived from the DEC forms:

$$A_{ij} = \text{Softmax}\left(\frac{Q_i K_j^\top}{\sqrt{d}} + W_{\text{conn}} \cdot \text{RBF}(D_{\text{DEC}}(i, j)) + W_{\text{curv}} \cdot \text{Tr}(\Omega_i \Omega_j^\top)\right), \quad (3)$$

where $D_{\text{DEC}}(i, j) = \|(\exp(-\tau \mathbf{L}_1))_{ij}\|_F$ is a *connection heat-transport* affinity built from the normalised connection Laplacian $\mathbf{L}_1$ (its off-diagonal blocks are the relative frame rotations $E_i^\top E_j$), and $\text{Tr}(\Omega_i \Omega_j^\top)$ is a curvature inner product. These biases give sparse visible tokens a gauge-invariant channel to communicate along the manifold.

Cartan–Darboux decoder. Each masked residue m is assigned a visible boundary anchor $a(m)$. The decoder predicts, per masked residue, its connection 1-form $\hat{\omega}_m$ and curvature 2-form $\hat{\Omega}_m$, together with the anchor-relative soldering form. Frames and coordinates are then recovered *analytically* by integrating the discrete Cartan structure equations along a path from the anchor,

$$\hat{E}_m = E_{a(m)} \exp\left(\sum_{e \in \text{path}(a(m) \rightarrow m)} \hat{\omega}_e\right), \quad \hat{t}_m = t_{a(m)} + \sum_{e=(u,v) \in \text{path}} \hat{E}_u \hat{\theta}_e. \quad (4)$$

Because the targets are gauge-invariant forms and reconstruction requires composing predictions along a path, the encoder cannot solve the task by local interpolation — it must localise masked geometry from sparse long-range context.

Mask-centric objective. The training loss sums differential-form reconstruction terms over masked residues: the soldering form ($\mathcal{L}_\theta$), the anchor-relative position and orientation ($\mathcal{L}_{\text{pos}}, \mathcal{L}_{\text{rot}}$), the connection 1-form ($\mathcal{L}_\omega$), and the curvature 2-form energy ($\mathcal{L}_\Omega$):

$$\mathcal{L} = \mathcal{L}_\theta + \mathcal{L}_{\text{pos}} + \mathcal{L}_{\text{rot}} + \mathcal{L}_\omega + \mathcal{L}_\Omega. \quad (5)$$

Restricting the loss to masked residues removes visible-token shortcuts. After pretraining the decoder is discarded and the encoder is frozen for downstream probing.

Table 1: Cartan-DEC vs. vanilla MAE at full corpus scale (369,740 structures, matched encoder / data / steps / probe). `acc`=top-1, `F1`=macro F1 over all 965 classes. Δ rows are relative to the vanilla-MAE baseline; *green*=improvement. The public MiAE-B reference row calibrates our frozen probe against published linear-probing numbers. **Best** among from-scratch rows in bold.

Pretraining condition	TEDBench test		CATH 4.4 test	
	acc	F1	acc	F1
Random-init encoder (no pretrain)	40.13	9.69	30.49	10.24
Vanilla-MAE (from scratch) [<i>baseline</i>]	56.91	32.23	51.10	23.90
Cartan-DEC-MiAE (DEC objective)	59.30	37.73	61.79	34.94
+ curvature-biased attention	60.18	39.30	64.12	37.93
+ holonomy-curvature masking	57.58	30.13	64.45	34.18
Δ vs. vanilla-MAE baseline				
Cartan-DEC (DEC objective)	+2.39	+5.50	+10.69	+11.03
+ curvature-biased attention	+3.27	+7.07	+13.02	+14.03
Reference (public MiAE-B, frozen, our probe)				
MiAE-B avg-pool (Hayes et al., 2025)	74.20	57.79	86.09	66.73
published MiAE-B (linear probe)	72.94	58.52	85.31	66.18

4 EXPERIMENTS

4.1 SETUP

We evaluate on the 965-class CATH topology (T-level) fold-classification task of TEDBench (Lau et al., 2024; Sillitoe et al., 2021), reporting top-1 accuracy (`acc`) and macro F1 over all 965 classes (`F1`) on both the in-distribution TEDBench test set (46,218 structures) and the external CATH 4.4 set (27,638 structures). Macro F1 is the primary metric given the extreme class imbalance. Unless stated otherwise, every encoder is a 12-layer, 512-dim (~ 38 M-parameter) transformer paired with a 3-layer, 256-dim Cartan–Darboux decoder, pretrained *from scratch* with AdamW (Loshchilov & Hutter, 2019) and a one-cycle schedule. All models are evaluated with an *identical* frozen-encoder probe: mean+max-pooled token latents feed a shared MLP head trained with label smoothing 0.1, so any downstream difference reflects only the learned encoder. Deterministic seeds and cached geometry make every number reproducible. Additional details are in Appendix A.

4.2 CARTAN-DEC VS. VANILLA MAE AT FULL CORPUS SCALE

Our headline experiment holds the architecture, data, step count, and probe fixed and varies only the pretraining objective, on the *full* 369,740-structure TEDBench-train corpus. Table 1 and Figure 2 report the results. Replacing the vanilla-MAE soldering target with the Cartan-DEC differential-form objective improves top-1 accuracy by +2.39 (TED) and +10.69 (CATH), and macro F1 by +5.50 and +11.03 — a large gain out-of-distribution, where global structural reasoning matters most. Bi-asing the encoder attention with curvature and connection-heat priors adds further improvements, reaching +3.27 (TED) / +13.02 (CATH) top-1 over vanilla MAE. Pretraining itself is essential: the Cartan-DEC encoder beats a random-init encoder by +19.2 (TED) / +31.3 (CATH) top-1.

Two observations temper the story. First, holonomy-curvature masking (the “full idea”) helps external CATH accuracy (+13.36 top-1 vs. vanilla) but *hurts* in-distribution macro F1, a tension we dissect in §4.3. Second, our from-scratch encoders are smaller than, and pretrained on less data than, the public MiAE-B, so absolute numbers sit below the fully pretrained model; the scientific claim is the *controlled delta* of the objective, measured at full corpus scale. The reference row confirms the probe is faithful: the public MiAE-B encoder under our exact pipeline scores 74.2/57.8 (TED) and 86.1/66.7 (CATH), matching its published linear-probing numbers.

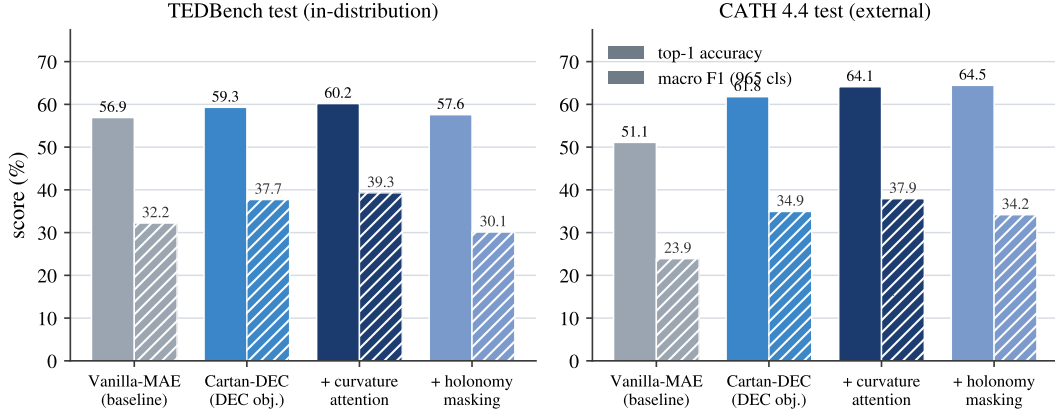


Figure 2: Full-scale results. Top-1 accuracy (solid) and macro F1 (hatched) for the four from-scratch pre-training conditions on the in-distribution (left) and external (right) test sets. The Cartan-DEC objective and curvature-biased attention improve substantially over vanilla MAE, especially out-of-distribution; holonomy-curvature masking helps external accuracy but hurts in-distribution F1 (analysed in §4.3).

Table 2: Ablation 1: loss recipes. Which differential form targets each condition optimises (soldering θ is al-acc vs. the no-bias encoder. Combining curvature and ways on).

Condition	θ	pos		
		+rot	ω	Ω
Solder-only (vanilla)	✓			
No-curv	✓	✓	✓	
No-conn	✓	✓		✓
No-anchor	✓		✓	✓
Full DEC	✓	✓	✓	✓

Attention bias	Δ TED	Δ CATH	entropy
No bias (ref)	—	—	3.61
RBF (Euclid. dist)	+1.88	+3.43	2.67
Heat ($\tau=0.5$)	+1.65	+2.08	2.69
Curv only	−0.33	−2.15	3.48
Combined	+5.75	+6.33	3.04

4.3 ABLATIONS

We now vary one design factor at a time (fixed encoder, data, steps, and probe) to identify what drives the gains. Ablations 1–3 (loss terms, attention bias, masking) are in the main text; graph construction and decoder integration are in Appendix B.

Which loss terms matter? (Ablation 1) We pretrain five encoders differing only in which DEC targets enter the loss (Table 2, Figure 3a). The full Cartan-DEC recipe improves over the soldering-only vanilla-MAE baseline by +3.48 (TED) / +7.44 (CATH) top-1 and +3.53/+4.67 macro F1. A leave-one-out analysis (row – full recipe) identifies the **curvature 2-form energy Ω as the most valuable single term**: removing it is the only change that *hurts* both accuracy and F1 on both test sets (−1.53 CATH acc, −0.80 CATH F1). Curvature supervision provides the second-order signal that stabilises the decoder’s path integration; the anchor and connection terms are individually more redundant but jointly necessary to reach the full recipe.

Does geometry-aware attention help? (Ablation 2) Holding the full Cartan-DEC objective fixed, we vary only the additive attention logit bias (Table 3, Figure 3b). Every bias except the curvature-only variant beats the no-bias encoder, and the **combined connection-heat ($\tau=0.5$) + curvature bias is clearly best** (+5.75 TED / +6.33 CATH top-1) — markedly more than either channel alone, indicating the two priors are complementary. Per-layer attention entropy explains why: under the sparse masked regime the no-bias encoder collapses to low entropy in deep layers, the curvature-only bias barely changes this, but the combined bias maintains a healthy entropy (3.04 nats mean) that keeps information flowing among the few visible tokens.

Should masking target holonomy defects? (Ablation 3) A central hypothesis of the original idea was that masking regions of high holonomy defect would force better global reasoning. We test nine

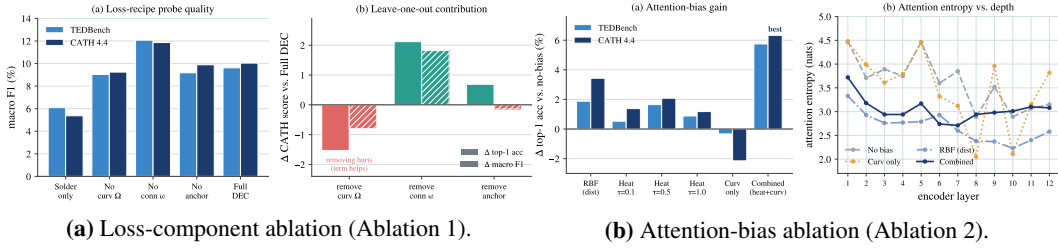


Figure 3: Dissecting the geometric components. (a) Macro F1 per loss recipe and the leave-one-out contribution of each term (negative Δ = removing the term hurts). (b) Per-variant Δ accuracy vs. a no-bias encoder, and per-layer attention entropy: the combined curvature+heat bias is strongest downstream while keeping attention from collapsing onto a few visible tokens.

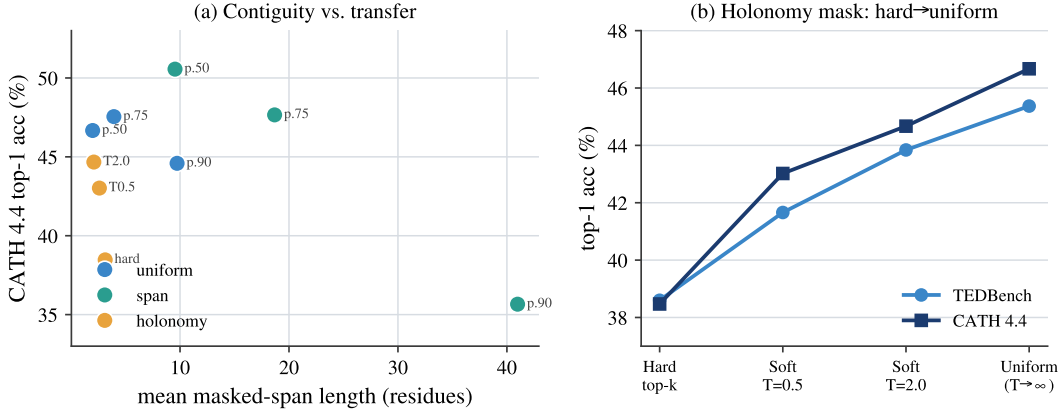


Figure 4: Masking ablation (Ablation 3). (a) External accuracy vs. mean masked-span length: contiguous span masking (green) dominates, while holonomy-based masking (amber) is scattered and weak. (b) Along the holonomy family, softening the sampling temperature moves performance monotonically from hard top- k up toward the uniform limit — soft > hard, but no holonomy variant beats uniform.

masking strategies at fixed everything-else (Figure 4). The result is a clean negative: **contiguity, not holonomy targeting, is the useful axis**. Moderate contiguous span masking ($p=0.5$) is the best condition on both benchmarks (+3.89 CATH top-1 vs. uniform), while hard holonomy top- k masking is the *worst* holonomy variant (−8.20 CATH top-1), starving the encoder of the global anchor context it needs. Soft Boltzmann sampling recovers *monotonically toward uniform* as temperature rises (Figure 4b): its high-temperature limit undoes the damage of hard targeting but never exceeds plain uniform masking. The holonomy-defect prior thus carries no net downstream benefit at this scale; its value is bounded above by the uniform limit it converges to. We report this faithfully as a case where a geometrically appealing idea does not pay off.

Reconstruction fidelity anti-correlates with transfer. An ablation over the Cartan–Darboux integration pathway (Appendix B, Figure 5b) surfaces a striking effect. Multi-hop MST path integration achieves by far the lowest masked-residue reconstruction RMSD (2.61 Å) yet the *worst* downstream transfer (−3.12 CATH top-1 vs. the 1-hop reference), whereas the strongest downstream encoder (K-shortest multi-path) reconstructs *less* accurately. Attaching each masked residue to a nearby parent lets the decoder solve reconstruction locally and relieves the encoder of the long-range localisation that actually transfers. **Harder reconstruction targets make better encoders** — a design principle that also explains why aggressive masking and gauge-invariant (anchor-relative) targets help.

4.4 DISCUSSION

Taken together, the experiments support a clear message and two useful cautions. The message: supervising the intrinsic differential geometry of a protein — its connection and curvature forms — is a stronger self-supervision signal than reconstructing coordinates alone, and a curvature/heat-

transport attention prior compounds the gain. The cautions: the more elaborate geometric heuristics (holonomy-targeted masking, deep path integration) optimise the wrong quantity, and reconstruction quality is not a reliable proxy for representation quality. These negative results are, we believe, as informative as the positive ones for future work on geometry-aware structural pretraining.

5 CONCLUSION

We introduced Cartan-DEC-MiAE, which recasts masked protein-structure autoencoding in the language of Cartan moving frames and discrete exterior calculus. By predicting and integrating gauge-invariant connection and curvature forms, and by biasing sparse-encoder attention with curvature and connection-heat priors, Cartan-DEC-MiAE learns markedly better fold representations than an identically trained vanilla MAE at full corpus scale, particularly out-of-distribution. Controlled ablations pinpoint curvature 2-form supervision and combined attention biases as the key ingredients, while cautioning that holonomy-targeted masking and low-RMSD reconstruction are counter-productive. We hope these results, positive and negative alike, encourage the community to treat discrete differential geometry as a first-class inductive bias for structural self-supervision. Promising directions include end-to-end fine-tuning of the Cartan-DEC encoder, scaling to the full AFDB, and extending the differential-form vocabulary to all-atom and multi-chain geometry.

REPRODUCIBILITY STATEMENT

All reported numbers use fixed seeds (train seed 1, eval seed 42), deterministic data subsets, and cached geometry; the experimental protocol, architecture, and metric definitions are specified in §4.1 and Appendix A, and every ablation isolates a single factor with all others held fixed.

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A IMPLEMENTATION DETAILS

Geometry featurisation. Contact graphs use an 8 Å C_α ball (1-skeleton) with contact triangles as the 2-skeleton; connection 1-forms, curvature 2-forms, holonomy defects, and connection heat kernels are precomputed and cached, and reused verbatim across all conditions. On real, frame-exact coordinates the discrete torsion and the true group holonomy vanish identically, so the learnable targets are the connection 1-forms ω , curvature 2-forms Ω , and the integrated Cartan–Darboux anchor geometry (non-trivial because $\log$ on $SO(3)$ is non-linear).

Architecture and optimisation. Encoder: 12 layers, dim 512, 8 heads (~ 38 M parameters). Cartan–Darboux decoder: 3 layers, dim 256, discarded after pretraining. Optimiser: AdamW (Loshchilov & Hutter, 2019) with one-cycle LR; mask ratio 0.5 (full-scale run) unless otherwise noted. Pretraining uses 5 epochs at full corpus scale and 10 epochs on the 50K/20K subsets used by the ablations.

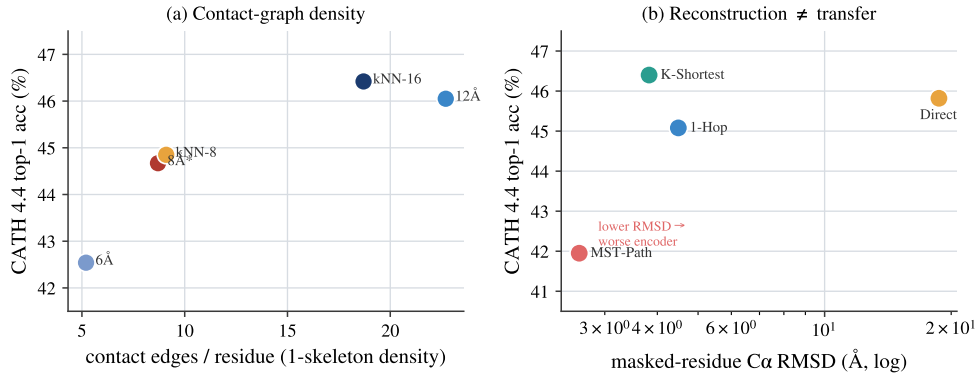
Probe. The frozen encoder produces mean+max-pooled token latents (1024-dim); an identical MLP head is trained for 50–100 epochs (AdamW, cosine, label smoothing 0.1), shared across all conditions so downstream differences reflect only the encoder. Evaluation covers the full 46,218 TEDBench-test and 27,638 CATH-4.4 structures, macro F1 over all 965 CATH-T classes.

Table 4: Ablation 4: contact-graph construction. e/res=contact edges per residue. 8 Å ball is the default.

Config	e/res	TED		CATH	
		acc	F1	acc	F1
Radius 6 Å	5.2	42.50	9.77	42.54	9.74
Radius 8 Å	8.7	43.76	9.86	44.67	9.96
Radius 12 Å	22.7	46.01	11.36	46.05	10.49
kNN $k=8$	9.1	44.19	10.03	44.85	10.51
kNN $k=16$	18.7	45.85	10.62	46.42	11.26

Table 5: Ablation 5: decoder integration. RMSD = masked-residue C α reconstruction error.

Condition	CATH acc	CATH F1	RMSD (Å)
Direct	45.82	11.04	18.70
1-Hop (ref)	45.08	9.93	4.49
MST-Path	41.95	8.02	2.61
K-Shortest	46.40	10.87	3.82

**Figure 5: Geometry and decoder ablations.** (a) Denser contact graphs (more edges/residue) give modestly better transfer; a 6 Å cutoff is too sparse. (b) Reconstruction fidelity anti-correlates with transfer: the lowest-RMSD decoder (MST-Path) yields the weakest encoder, while harder targets (K-shortest, Direct) transfer better.

B ADDITIONAL ABLATIONS

Contact-graph construction (Ablation 4). We vary the contact-graph cutoff and triangulation rule, holding model, objective, masking, and probe fixed (Table 4, Figure 5a). A 6 Å ball is too sparse (−2.12 CATH top-1 vs. the 8 Å default), thinning the 2-skeleton; denser graphs help modestly (12 Å: +1.38; kNN-16: +1.75 CATH top-1) at a small preprocessing cost. Downstream quality tracks 2-skeleton density more than the specific rule.

Decoder path integration (Ablation 5). We vary how the Cartan–Darboux decoder integrates predicted forms into geometry (Table 5, Figure 5b). K-shortest multi-path integration transfers best (+1.33 CATH top-1 vs. 1-hop), while multi-hop MST integration — despite the lowest reconstruction RMSD — transfers worst, the anti-correlation analysed in §4.3.

Full masking breakdown (Ablation 3). Table 6 lists all nine masking conditions with achieved mask ratio, mean masked-span length, and full metrics, complementing Figure 4.

Table 6: Ablation 3: full masking-strategy results (all 965 classes, full test sets). Contiguous span masking at a moderate ratio is best; no holonomy variant beats uniform masking.

Condition	mask p	span	TEDBench		CATH 4.4	
			acc	F1	acc	F1
Uniform $p=0.50$	0.50	1.98	45.37	10.46	46.67	11.34
Uniform $p=0.75$	0.75	3.93	45.92	9.95	47.55	11.71
Uniform $p=0.90$	0.90	9.73	43.51	8.20	44.59	9.70
Span $p=0.50$	0.50	9.54	47.41	11.22	50.56	13.44
Span $p=0.75$	0.75	18.68	46.38	11.84	47.66	12.50
Span $p=0.90$	0.90	40.98	36.37	4.73	35.66	5.35
Hard-holonomy	0.50	3.13	38.60	5.52	38.47	6.14
Soft-holonomy $T=0.5$	0.50	2.60	41.66	7.54	43.02	8.90
Soft-holonomy $T=2.0$	0.50	2.09	43.84	8.42	44.67	9.26